Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Department of Computer Science and Engineering

MST Examination (Summer Semester-2025)

B.E.
29th June 2025
Duration: 2 Hours
Max Marks: 30

Course Code: UCS414/UCS520
Course Name: Computer Networks
Name of Faculty: Dr. Sumit Kumar

Note: Attempt all questions. Assume missing data, if any, suitably.

Q1	(a) Differentiate between the following types of communication modes in data transmission with suitable examples: simplex and full-duplex. (b) Explain the concept and purpose of a Storage Area Network (SAN). (c) Describe the advantages of the mesh topology with a suitable example. If there are 20 devices in the network, then calculate the total number of point-to-point links required. Also, determine how many links each device will have.	(2) (2) (2)
Q2	(a) State any four services provided by the Transport Layer in the OSI reference model. (b) Analyse the consequences of the looping problem on network performance in bridged networks. Suggest a suitable mechanism that is used to solve the looping problem. (c) Illustrate the frame format of standard Ethernet (IEEE 802.3). List its fields in order and briefly explain the purpose of each field.	(2) (2) (2)
Q3	Assume that the previous signal level was positive, then draw the signal pattern for the data 00110101 using the line coding techniques mentioned below. i) Unipolar NRZ ii) NRZ-I iii) Polar RZ iv) Differential Manchester v) Bi-Polar AMI vi) 2B1Q	(6)
Q4	(a) Discuss the purpose of the Address Resolution Protocol (ARP). Draw the ARP frame format and briefly explain the function of each field. (b) A sender is using the Go-Back-N ARQ protocol with a 3-bit sequence number field. Assume both sender and receiver are initially expecting sequence number 0. The following events occur in sequence: i. Frame 0 is sent and acknowledged. ii. Frames 1 and 2 are sent. iii. ACK for Frame 1 is received. iv. ACK for Frame 2 is lost. v. Timeout occurs for Frame 2.	(3) (3)

	vi. Sender retransmits Frame 2 and received acknowledgment, then sends Frames 3 and 4. After each event, determine the Sender's window (i.e., the range of sequence numbers it can send). Also, after each event, determine the Receiver's window (i.e., the expected in-order sequence number).	
Q5	(a) Draw and explain the flowchart representing the CSMA (Carrier Sense Multiple Access) persistence methods: 1-persistent and p-persistent CSMA. (b) A sender wants to transmit the following four 8-bit binary data words using the checksum method. Word 1: 10110101 Word 2: 11001100 Word 3: 11110000 Word 4: 00011101 Calculate the checksum at the sender side and prepare the complete data (original words + checksum) to be sent. At the receiver side, assume the same data (all 5 words, including checksum) is received correctly. Perform the checksum verification process and state whether the data is accepted or rejected.	(3) (3)

---------- END ----------